S.P.E.C.T.R.E. Major Project Presentation Script

Secure Portable Encrypted Communication Terminal for Remote Environments

Department of Electronics and Communication Engineering, B.M.S. College of Engineering, Bengaluru

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Project & Team Overview

- **Project Title:** S.P.E.C.T.R.E. (Secure Portable Encrypted Communication Terminal for Remote Environments)
- **Project Guide:** Dr. Arathi R Shankar, Associate Professor, Dept. of ECE, BMSCE
- **Head of Department:** Dr. Beena Ullala Mata B N, Professor & Head, Dept. of ECE, BMSCE

Speaker Role Allocation & Division

Speaker	Full Name	USN	Core Allocated Domain	Assigned Slides
Speaker 1	**Maneesh S M**	1BM23EC139	**Introduction, Threat Paradigm, Problem Formulation & Hardware Architecture**	Slides 01 – 06
Speaker 2	**Krishna Gupta**	1BM23EC123	**Physical Layer, FreeRTOS Compute Subsystem, COMSEC Cryptography & DTN Mesh**	Slides 07 – 11
Speaker 3	**Dhruv Kumar Koshta**	1BM23EC078	**C2 Gateway, Full-Stack Battlespace Dashboard, & Remote Anti-Tamper Zeroization**	Slides 13 – 15
Speaker 4	**Kshitij D**	1BM23EC124	**Edge-AI EW Detection, Biomechanical Casualty Sensing, Empirical Results, Benchmarks & Roadmap**	Slides 12, 16 – 22

SECTION 1: INTRODUCTION, PROBLEM FORMULATION & HARDWARE OVERVIEW

Presenter: Maneesh S M (1BM23EC139)

(Slides 01 to 06)

Slide 01: Title Slide & Project Introduction

SLIDE VISUAL : Title Slide — S.P.E.C.T.R.E., Project Meta, Team Names, College Emblems, and Hardware Prototype Photo]*

Maneesh S M:

"Respected Head of the Department Dr. Beena Ullala Mata ma'am, our project guide Dr. Arathi R Shankar ma'am, and esteemed members of the evaluation committee. A very warm good morning to all of you.

I am **Maneesh S M**, and on behalf of my team members **Krishna Gupta**, **Dhruv Kumar Koshta**, and **Kshitij D**, I take immense pride in presenting our major project work titled:

'S.P.E.C.T.R.E. — Secure Portable Encrypted Communication Terminal for Remote Environments.'

In one sentence: S.P.E.C.T.R.E. is an infrastructure-independent, sovereign, low-SWaP (Size, Weight, and Power) tactical mesh communications ecosystem. Built on commercial off-the-shelf silicon, it delivers authenticated AES-256-GCM tactical messaging, asynchronous frequency-hopping spread spectrum, non-volatile delay-tolerant networking, on-chip machine learning jamming detection, and authenticated remote zeroization — operating completely off-grid without dependence on cellular towers, satellites, or GPS constellations."

Slide 02: Operational Context & Battlefield Failure Modes

SLIDE VISUAL : Tactical theater breakdown — Hostile Electronic Attack, Emitter Geolocation (RDF), Capture & Key Compromise]*

Maneesh S M:

> "Let us examine why this project is critical. In modern Network-Centric Warfare (NCW) and emergency disaster operations, tactical units live and die by the Boyd OODA loop — *Observe, Orient, Decide, and Act*. Tactical commanders require continuous Command and Control (C2) situational awareness.
>
> However, forward echelons systematically lose communications at the exact moment they need it most. Contemporary tactical networks rely on three fragile pillars: 1. **Commercial & Tactical Cellular (LTE/5G):** Completely centralized; a single kinetic strike on a cellular tower or fiber severance isolates the entire detachment. 2. **Satellite Communications (SATCOM):** Highly vulnerable to orbital jamming, high-power uplink noise, and satellite denial in contested Anti-Access/Area Denial (A2/AD) theaters. 3. **High-Power VHF/UHF Voice Radios (e.g., SINCGARS):** They transmit continuous-wave voice for 3 to 15 seconds at 5 to 50 Watts of power.
>
> This creates three fatal battlefield failure modes: - **Hostile Electronic Warfare (EW):** Barrage and spot jammers saturate the spectrum and sever narrowband links. - **Signal Intelligence (SIGINT) & Radio Direction Finding (RDF):** Long continuous voice transmissions give adversary phase-interferometric arrays the 150 to 500 ms dwell time required to calculate the Angle of Arrival, triangulating soldier positions with precision artillery strikes. - **Terminal Capture & Key Compromise:** When an operator is captured, static symmetric keys stored in flash memory expose the entire communication echelon to decryption."

Slide 03: Problem Formulation & The Survivability Model

SLIDE VISUAL : Mathematical Survivability Model equation, risk factors, and comparison of conventional options]*

Maneesh S M:

> "To address these failure modes systematically, we formulated a qualitative tactical risk model where force survivability $P_{survive}$ is expressed as the product of three independent operational probabilities:
>
> $$P_{survive} \propto A_{C2} \times (1 - P_{detect}) \times (1 - P_{exploit})$$
>
> Where: - **A_{C2}** is tactical link availability under jamming and physical terrain masking. We maximize this through LoRa Chirp Spread Spectrum processing gain, multi-hop mesh flooding, and Delay-Tolerant Networking (DTN). - **P_{detect}** is the probability of adversary electromagnetic intercept. We drive this down by compressing our encrypted messages into sub-100-millisecond bursts (SF7 / 250 kHz bandwidth) paired with pseudo-random frequency hopping. - **$P_{exploit}$** is the probability that captured hardware yields cryptographic keys or plaintext intelligence. We drive this to zero through zero-trust authenticated encryption on every frame and remote one-touch hardware zeroization.
>
> Conventional options fail: military radios like the Harris Falcon III cost upwards of \$15,000 USD and remain export-controlled; cellular MANETs die without infrastructure; and civilian mesh platforms like Meshtastic lack authenticated dynamic keying, electronic warfare detection, and hardware anti-tamper security."

Slide 04: Proposed Solution — 5-Tier Defense-in-Depth Architecture

SLIDE VISUAL : Five-tier block diagram — Tier 1 PHY to Tier 5 C2 with functional data flow]*

Maneesh S M:

"To solve this tri-fold problem, S.P.E.C.T.R.E. implements an integrated Five-Tier Defense-in-Depth Architecture: - **Tier 1 (Physical Layer):** Semtech SX1278 transceiver operating at 433 MHz UHF utilizing Chirp Spread Spectrum modulation for high processing gain below the noise floor. - **Tier 2 (Compute & Task Isolation):** Espressif dual-core 240 MHz microcontroller running FreeRTOS with strict hardware-level core affinity. - **Tier 3 (Communications Security - COMSEC):** Cryptographically authenticated AES-256-GCM encryption with 96-bit random IVs and 128-bit GMAC integrity tags on every frame. - **Tier 4 (Resilient Mesh & Edge-AI):** Asynchronous CAD-driven Frequency Hopping (FHSS), non-volatile SPIFFS Delay-Tolerant Networking, and on-chip 100-tree Isolation Forest jamming classification. - **Tier 5 (Command & Control):** A headless gateway coupled over an air-gapped USB serial bridge to our Tactical Command Center (TCC) dashboard."

Slide 05: Hardware Architecture & Subsystems

SLIDE VISUAL : Component block diagram, breadboard prototype photos with pin mapping and callouts]*

Maneesh S M:

"Here is our actual physical hardware implementation. The terminal is built around the dual-core **ESP32-WROOM-32** microcontroller clocked at 240 MHz with 520 KB SRAM, hardware RNG, and cryptographic accelerators.

The hardware architecture comprises six tightly coupled subsystems: 1. **Compute Subsystem:** Dual-core Xtensa running FreeRTOS with non-volatile flash storage. 2. **RF Frontend:** Semtech SX1278 transceiver interfaced via a 10 MHz SPI bus with hardware interrupt lines DIO0 (packet Rx/Tx done) and DIO1 (CAD detection), connected to a matched 433 MHz helical antenna. 3. **Tactical Input Matrix:** A 16-key (4×4) sealed mechanical switch matrix debounced at 50 ms, allowing instant single-click dispatch of 9 standardized Indian military tactical prowords (e.g., CONTACT, SATHI GHAYAL, MAYDAY, LZ HOT). 4. **Visual Display:** A 0.96-inch monochrome SSD1306 OLED over a 400 kHz I2C bus with zero backlight leakage for night-vision discipline. 5. **Power Management Unit (PMU):** 18650 rechargeable Li-Ion cell with an AMS1117 3.3V LDO regulator and multi-stage bulk tantalum decoupling (470 µF) to prevent brownout resets during high-power +17 dBm transmit pulses. 6. **Dedicated C2 Gateway Unit:** A headless unit that strips UI rendering to dedicate 100% of compute cycles to high-speed RF ingestion and streaming telemetry."

Slide 06: Sprint-Based Engineering Execution

SLIDE VISUAL : Sprint roadmap timeline from Sprint A to Sprint H/I with completion status chips]*

Maneesh S M:

> "To build a defense-grade cyber-physical system within our academic timeline, we executed our engineering through a rigorous, sprint-based delivery cadence: - **Sprint A:** RF transceiver bring-up, SPI interrupt orchestration, and AES-256-GCM framing. - **Sprint B:** Multi-hop TTL mesh routing and the 15-channel asynchronous CAD-FHSS engine. - **Sprint C:** Non-volatile SPIFFS DTN store-carry-forward engine and 4×4 keypad tactical prowords. - **Sprint D:** Isolation Forest model training, C-header compilation, and remote zeroization. - **Sprint E:** Passive RF path-loss distance estimation and tactical sector radials. - **Sprint G & H:** 5-stage cryptographic pipeline inspector and real-time electronic warfare injection testbench.
>
> Having established the problem, threat landscape, and hardware foundation, I now hand over the presentation to **Krishna Gupta**, who will detail the physical layer, deterministic FreeRTOS compute architecture, cryptographic pipeline, and delay-tolerant mesh networking."

SECTION 2: PHYSICAL LAYER, FREERTOS COMPUTING, COMSEC & MESH

Presenter: Krishna Gupta (1BM23EC123)

(Slides 07 to 11)

Slide 07: Physical Layer Implementation & LPD Burst Timing

SLIDE VISUAL : LoRa chirp spectrogram, Shannon-Hartley formulation, Time-on-Air timing breakdown, and Link Budget]*

Krishna Gupta:

"Thank you, Maneesh. Respected evaluators, I will walk you through the embedded software runtime, the physical RF modulation, our zero-trust cryptographic implementation, and the delay-tolerant mesh network.

To survive against near-peer Signal Intelligence (SIGINT), S.P.E.C.T.R.E. rejects legacy narrowband voice radios in favor of **Chirp Spread Spectrum (CSS)** modulation operating at 433 MHz.

According to the **Shannon-Hartley Theorem**: $$C = B \cdot \log_2\left(1 + \frac{S}{N}\right)$$ By spreading our signal across a wide 250 kHz bandwidth with Spreading Factor SF7, our system achieves a processing gain of **16.6 dB**: $$PG = 10 \cdot \log_{10}\left(\frac{250,000}{5,468}\right) \approx 16.6\text{ dB}$$ This allows the SX1278 transceiver to decode tactical packets arriving up to **7.5 dB below the thermal noise floor**, forcing adversary jammers to broadcast immense power density across 250 kHz to disrupt our link.

Low Probability of Detection (LPD) Timing: Adversary Radio Direction Finding (RDF) phase-interferometers require a minimum dwell time of 150 to 500 milliseconds to triangulate a radio's Angle of Arrival (AoA). Using Semtech's physical-layer formulation, our symbol duration at SF7 is exactly **0.512 ms**: - Preamble duration (12.25 symbols): **6.272 ms** - Payload duration (160 symbols for our encrypted packet): **81.920 ms** - **Total Time-on-Air (ToA):** $6.272 + 81.920 = \mathbf{88.192\text{ ms}}$

Because our burst finishes in under 100 milliseconds, adversary sweeping spectrum analyzers complete their sweep without detecting the carrier, driving the probability of intercept down by over 94%. With +17 dBm transmit power, our Maximum Allowable Path Loss is **138.5 dB (up to 142.5 dB with boosted PA)**, yielding over 2.2 km range in dense concrete urban clutter and over 8 km in open terrain."

Slide 08: Dual-Core Compute & Deterministic FreeRTOS Task Isolation

SLIDE VISUAL : Dual-core architecture diagram — Core 0 (PRO_CPU) vs Core 1 (APP_CPU) with FreeRTOS queues]*

Krishna Gupta:

> "A fatal vulnerability in conventional embedded tactical radios is **cryptographic starvation**: when intensive 256-bit bignum math or radio polling runs on a single microcontroller thread, the user interface freezes, button presses are dropped, and critical RF reception windows are missed.
>
> S.P.E.C.T.R.E. resolves this by leveraging the Symmetric Multiprocessing (SMP) capabilities of FreeRTOS on the ESP32, enforcing strict hardware-level task isolation: - **Core 0 (PRO_CPU — Radio & Cryptographic Subsystem):** Runs `TaskRadio` at high priority (Priority 3, 48 KB stack). Core 0 is dedicated exclusively to servicing SX1278 hardware interrupts (DIO0 on GPIO 26 and DIO1 on GPIO 35), executing mbedTLS bignum operations, authenticated AES-256-GCM encryption/decryption, jittered heartbeats (14–17s), and DTN checks. - **Core 1 (APP_CPU — Operator UI & Sensor Subsystem):** Runs the main Arduino task at Priority 1. It executes the 50 ms debounce matrix keypad scan every 5 ms, drives the tactical menu state machine, updates the SSD1306 OLED display, and handles power monitoring.
>
> **Inter-Core Communication:** Core 0 and Core 1 communicate exclusively through thread-safe, non-blocking FreeRTOS queues — `txQueue` and `rxQueue` — configured with depth 5 storing `MessageEvent` structs copy-by-value. The UI thread never touches the SPI bus, ensuring zero UI stutter and zero dropped RF frames."

Slide 09: Communications Security (COMSEC) Architecture

`SLIDE VISUAL` : 48-byte packet format diagram, AES-256-GCM authenticated cipher pipeline, and 1-bit tamper proof math]*

Krishna Gupta:

> "In hostile operational environments, pre-shared static keys are a fatal vulnerability. If a single radio is captured, all communications are blown. S.P.E.C.T.R.E. enforces zero-trust COMSEC:
>
> **Over-the-Air Data Frame Structure (48-byte header + ciphertext):** Every frame carries: - Magic Byte `0xAB` (Protocol Identifier) - Message Type ID (1 Byte), Hop Count (1 Byte), Payload Length (1 Byte) - Originator Node ID (16 Bytes, e.g., `Alpha-1`) - Cryptographically random Initialization Vector (12 Bytes / 96-bit IV) - Galois Authentication Tag (16 Bytes / 128-bit GMAC Tag) - AES-256-GCM Ciphertext (Variable up to 127 Bytes)
>
> **Galois/Counter Mode (GCM) Integrity:** For every single packet, the hardware True Random Number Generator generates a fresh 96-bit IV. Encryption occurs in counter mode, while a 128-bit authentication tag T_{auth} is calculated using polynomial evaluation over the binary Galois field $GF(2^{128})$: $$T_{auth} = \text{GHASH}_H(A \parallel C \parallel \ell(A) \parallel \ell(C)) \oplus \text{AES}_K(\text{CB}_0)$$
>
> In legacy stream ciphers (like AES-CTR), an adversary can intercept a packet and flip coordinate bits (directing friendly fire) because XOR is linear. In S.P.E.C.T.R.E., flipping even a single bit in the ciphertext causes the Galois field tag verification to fail with probability $1 - 2^{-128}$. The packet is instantly dropped at the edge with zero plaintext leakage."

Slide 10: Electronic Counter-Countermeasures (ECCM) — Asynchronous CAD-FHSS

SLIDE VISUAL : 15-channel allocation table, 37-symbol sync strobe timing diagram, and CAD sweep state machine]*

Krishna Gupta:

"To defeat hostile tracking and high-power carrier jammers, S.P.E.C.T.R.E. implements packet-level Frequency Hopping Spread Spectrum across 15 orthogonal channels (433.050 MHz to 437.250 MHz with 300 kHz channel spacing).

Solving the Decentralized Rendezvous Problem without GPS: Traditional frequency-hopping radios require synchronized atomic clocks or GPS time signals. But in contested theaters, GPS is the first thing adversaries jam or spoof. How do two unsynchronized nodes rendezvous without GPS?

We engineered an **Asynchronous Channel Activity Detection (CAD) Sweep State Machine**: 1. **Receiver CAD Sweep:** When idle, the receiver rapidly sweeps all 15 channels. On each channel, it performs a 1.174 ms CAD energy sample on DIO1. The full 15-channel sweep completes in: $$T_{sweep} = 15 \times 1.174\text{ ms} = \mathbf{17.610\text{ ms}}$$ 2. **Transmitter Preamble Strobe:** When a node initiates transmission, it prepends an extended 37-symbol preamble strobe: $$T_{strobe} = 37 \times 0.512\text{ ms} = \mathbf{18.944\text{ ms}}$$ 3. **Mathematical Proof of Rendezvous:** Because $T_{strobe} > T_{sweep}$ ($18.944\text{ ms} > 17.610\text{ ms}$), the transmitter's strobe is **mathematically guaranteed to intersect** the scanning receiver's CAD detection window regardless of clock drift or phase offset! Upon detection, the receiver locks onto the carrier and ingests the authenticated payload."

Slide 11: Multi-Hop Mesh & Delay-Tolerant Networking (DTN)

SLIDE VISUAL : Five-step DTN storyboard: Transmit → Relay → Link Blackout → Flash Store → Data Mule Dump]*

Krishna Gupta:

> "In tactical maneuvers across deep ravines, urban concrete canyons, or subterranean bunkers, radio links fragment. Conventional ad-hoc networks simply drop undelivered packets, losing vital reconnaissance data.
>
> S.P.E.C.T.R.E. integrates an autonomous **Delay-Tolerant Networking (DTN)** store-carry-forward engine: 1. **TTL Flooding:** Standard tactical messages are transmitted with a Time-to-Live of 3 hops. Intermediate nodes decrypt the header, decrement the hop count, apply a randomized jittered back-off (50 to 250 ms) to prevent RF collisions, and retransmit. 2. **Partition Detection:** Field nodes maintain a local presence table. If an acknowledgment is not received after 3 retries or a peer node is not heard for 30 seconds, the link is classified as partitioned. 3. **Non-Volatile Flash Buffering:** Instead of dropping the packet, Core 0 intercepts it and commits the raw ciphertext, IV, destination callsign, and timestamp directly into the ESP32's onboard SPIFFS flash memory queue (`/dtn_0001.bin` to `/dtn_0032.bin`). The ciphertext is never stored unencrypted, preserving COMSEC at rest. 4. **Autonomous Data Muling:** When the dismounted patrol physically maneuvers back into radio range of the squad or the C2 Gateway, the background heartbeat engine detects peer presence. The DTN state machine awakens, drains the buffered flash queue in controlled bursts (1 packet per 5 seconds), and securely unlinks the files upon confirmed delivery.
>
> I now hand over the presentation to **Dhruv Kumar Koshta**, who architected the C2 Gateway firmware, the high-speed serial telemetry protocol, our full-stack Tactical Command Center dashboard, and the remote zeroization kill-switch."

SECTION 3: C2 GATEWAY, FULL-STACK DASHBOARD & ZEROIZATION

Presenter: Dhruv Kumar Koshta (1BM23EC078)

(Slides 13, 14, 15)

Slide 13: C2 Gateway Firmware & Serial Telemetry Protocol

SLIDE VISUAL : C2 Gateway hardware, JSON telemetry schema listing, multi-node session key table, and serial bridge diagram]*

Dhruv Kumar Koshta:

"Thank you, Krishna. Respected evaluators, while forward operators carry the ruggedized field terminal, tactical commanders require real-time, air-gapped Command and Control (C2) situational awareness.

To eliminate the vulnerability of connecting command laptops directly to RF or IP networks, our architecture introduces a physical **Headless C2 Gateway (`spectre-c2-gateway`)**: - It is an air-gapped hardware firewall running on a dedicated dual-core ESP32. It has no screen, keyboard, or operating system — dedicating 100% of its processor cycles to high-throughput RF reception, hardware register extraction, and AES-256-GCM decryption. - **Multi-Node Session Key Table:** The gateway firmware maintains an active, thread-safe session table supporting up to 8 independent field terminals concurrently (e.g., `Alpha-1`, `Bravo-2`), tracking their session keys, heartbeat timestamps, and packet sequence counters with zero key collisions. - **USB UART JSON Telemetry Stream:** The gateway bridges decrypted data over a dedicated, isolated USB COM port running at 115,200 baud (8-N-1). Every received RF frame is serialized into a newline-delimited JSON document containing: - Node ID, callsign, and operational state (`ACTIVE`, `DEGRADED`, `JAMMED`, `LOST`) - Physical RF metrics: RSSI, SNR, and estimated path-loss distance - Full cryptographic diagnostics: Algorithm, 12-byte IV, 16-byte GMAC tag, and `tagMatch` status - 8-dimensional ML feature vectors and real-time Isolation Forest anomaly scores.

Bidirectional Host Command Ingestion: Tactical commanders can dispatch high-priority control strings down to the gateway using our rigid tokenized protocol: `CMD:::`. The gateway parses and queues these commands into its RF transmit engine within milliseconds."

Slide 14: Tactical Command Center (TCC) Situational Dashboard

SLIDE VISUAL : High-contrast TCC UI screenshot — 2D CRT Plan Position Indicator, 3D WebGL Battlespace Volume, Cryptographic Pipeline Inspector, and Active Node Roster]*

Dhruv Kumar Koshta:

> "To render this live telemetry stream for tactical decision-making, we designed and built the **Tactical Command Center (TCC) Dashboard (`spectre-dashboard`)**: - It is an enterprise-grade desktop application engineered using **React 18, TypeScript, Electron, Tailwind CSS, and Three.js (React-Three-Fiber)**, rendering fluid 60 FPS graphics with an authentic military green-phosphor CRT aesthetic.
>
> The dashboard provides four critical operational views: 1. **2D CRT Plan Position Indicator (PPI) Radar Scope:** - Simulates an authentic analog P31 cathode-ray phosphor sweep rotating at 1.5 rad/s with realistic radial luminance decay. - Features four calibrated concentric distance rings at 50 m, 100 m, 250 m, and 500 m. - When a packet arrives, a luminous green blip flashes at peak intensity at the node's calculated coordinates and decays exponentially over a 12-second half-life. 2. **3D WebGL Holographic Battlespace Volume:** - Built with Three.js, it projects a full 3D tactical volume with polar grids, elevation planes, tactical sector cones, and node beacons with vertical drop lines tethering them to the terrain plane — providing complete spatial awareness for multi-elevation or drone formations with 360-degree orbital camera controls. 3. **GPS-Denied Passive RF Distance & Sector Tracking:** - In A2/AD theaters where GPS is jammed, the gateway calculates distances using the empirical Log-Distance Path Loss model ($RSSI_0 = -42\text{ dBm}, n = 2.4$): $$d = 10^{\frac{-42.0 - \text{RSSI}}{24.0}}$$ - Distances are mapped onto standard fireteam operational sector radials (e.g., `Alpha-1` on 045° Northeast, `Bravo-2` on 225° Southwest), giving commanders instantaneous Blue Force Tracking without transmitting trackable downlink pings! 4. **Live Cryptographic Pipeline Inspector & 1-Bit Tamper Sandbox:** - A dedicated diagnostic console rendering all 5 sequential cryptographic stages of the incoming frame. Evaluators can click **'FLIP 1 BIT IN CIPHERTEXT'**; the dashboard immediately re-evaluates the Galois polynomial, displays a red cryptographic rejection banner, and confirms 100% Man-in-the-Middle mitigation."

Slide 15: Anti-Capture Security & Remote Hardware Zeroization

SLIDE VISUAL : Two-stage Slide-to-Arm dashboard trigger modal, zeroization sequence flowchart, and physical photo of OLED displaying ZEROIZED / KEYS DESTROYED]*

Dhruv Kumar Koshta:

> "In forward dismounted operations, equipment capture behind enemy lines is an inevitable risk. If an adversary captures a radio and uses laboratory probes or voltage glitching to read SRAM or flash, the entire operational echelon is compromised.
>
> S.P.E.C.T.R.E. implements an uncompromising, multi-tier **Anti-Tamper Cryptographic Zeroization Framework**: - **Tier 1 (Authenticated Remote C2 Kill-Switch):** When a forward node is captured or compromised, the commander selects the target on the TCC dashboard, slides the safety interlock to **ARM**, and clicks **INITIATE ZEROIZE**. The gateway broadcasts an authenticated `CMD:ZERO:` frame encrypted under the target's active session key. - **Tier 2 (Chassis-Breach Physical Panic Switch):** A physical microswitch along the enclosure seam is wired to GPIO 36 configured as a hardware Non-Maskable Interrupt (NMI). If the enclosure is opened by even 1.5 mm, the NMI preempts all CPU scheduling in under 200 nanoseconds.
>
> **Destructive 6-Stage Purge Routine:** Once triggered, the firmware executes a destructive emergency sequence: 1. **TRNG RAM Overwrite:** The hardware True Random Number Generator (`esp_fill_random`) overwrites all active 256-bit AES keys and ECDH contexts with random entropy noise, followed by an immediate `memset(0x00)` . 2. **mbedTLS Context Destruction:** All private key structures and bignum limb arrays are explicitly deallocated. 3. **Stack & Queue Sanitization:** FreeRTOS message queues and heap buffers are overwritten with zeroes. 4. **Non-Volatile Storage Flash Wipe:** `SPIFFS.format()` and `prefs.clear()` permanently erase all buffered DTN packets and operational history. 5. **OLED Visual Lockout:** Core 1 paints a permanent white-on-black lockout screen: *`ZEROIZED / KEYS DESTROYED / REFLASH REQUIRED`* . 6. **Permanent CPU Execution Halt:** The processor globally disables interrupts (`portDISABLE_INTERRUPTS()`) and vectors into a permanent `while(true)` halt loop — bricking the terminal into an inert piece of plastic.
>
> I now hand over the presentation to **Kshitij D**, who will present our Edge-AI jamming detection engine, biomechanical casualty sensing, empirical testbench results, comparative military benchmarks, and future roadmap."

SECTION 4: EDGE-AI EW DETECTION, BIOMECHANICS, RESULTS & CONCLUSION

Presenter: Kshitij D (1BM23EC124)

(Slides 12, 16 to 22)

Slide 12: Edge-AI Electronic Warfare Anomaly Detection

SLIDE VISUAL : 8-dimensional RF feature table, Isolation Forest decision tree traversal math, EWMA smoothing curve, and Threat Tier thresholds]*

Kshitij D:

> "Thank you, Dhruv. Respected evaluators, tactical radios in the field face sophisticated Electronic Warfare (EW): broadband barrage jamming, narrowband carrier spot jamming, and pulsed preamble disruption. Traditional military radios have no autonomous way to distinguish hostile jamming from natural terrain fading.
>
> S.P.E.C.T.R.E. resolves this by deploying an **on-chip Edge-AI Machine Learning Inference Engine** running directly on the ESP32 microcontroller:
>
> **Physical-Layer 8-Dimensional Feature Vector:** Rather than overwhelming the MCU with high-rate raw I/Q samples, during idle listening windows our firmware extracts eight engineered physical-layer RF features directly from the SX1278 registers: 1. Instantaneous RSSI (f_0) 2. Instantaneous SNR (f_1) 3. Wideband RF Noise Floor ($NF = \text{RSSI} - \text{SNR}$) ($f_2$) 4. Differential RSSI ($\Delta\text{RSSI}$) ($f_3$) 5. Differential Noise Floor (ΔNF) (f_4) 6. 10-packet Rolling Mean RSSI (μ_{RSSI}) (f_5) 7. 10-packet Rolling RSSI Standard Deviation (σ_{RSSI}) (f_6) 8. 10-packet Rolling Noise Floor Standard Deviation (σ_{NF}) (f_7)
>
> Rolling statistics are maintained via $O(1)$ circular ring buffers, requiring zero dynamic memory allocation.
>
> **100-Tree Embedded Isolation Forest:** We chose the Isolation Forest algorithm because it operates on an unsupervised isolation principle: anomalies isolate near the root of randomized decision trees with significantly shorter path lengths $h(x)$. We compiled 100 trained decision trees directly into static C++ arrays in `anomaly_engine.h`. The search traverses branch comparisons without expensive matrix multiplications, completing single-sample inference in merely **3.14 ms** on the 240 MHz Xtensa core!
>
> The raw score is smoothed using an Exponentially Weighted Moving Average (EWMA) filter ($\alpha = 0.30$): $$S_t = 0.30 \cdot s_{raw} + 0.70 \cdot S_{t-1}$$ Scores map to clear tactical threat tiers: - < 0.60: **NOMINAL (Clear Spectrum)** - $0.60 - 0.75$: **ELEVATED (Suspicious RF Activity)** - $0.75 - 0.85$: **HIGH (Spot Jamming Detected)** - ≥ 0.85: **CRITICAL (Barrage Jamming — Automated Frequency Hop Triggered)**"

[Technical Integration]: Biomechanical Casualty Detection Engine

SLIDE VISUAL : 3-Phase Casualty State Machine diagram: Kinetic Shock → Posture Collapse → Flaccid Atonia Variance Filter]*

Kshitij D:

> "Closely integrated with our edge intelligence is our **Inertial Biomechanical Casualty Detection Engine**. In infantry combat, incapacitated operators cannot press an SOS button. However, simple accelerometer threshold detectors fail catastrophically — firing a rifle generates 8g to 16g recoil spikes, and diving into cover produces 3g to 6g impacts.
>
> S.P.E.C.T.R.E. interfaces an InvenSense MPU-6050 6-DoF IMU sampled at 50 Hz on Core 1 to enforce a biologically invariant three-phase physiological state machine: 1. **Phase 1 (Kinetic Shock Detection):** Ballistic projectile strike triggers $|a| = \sqrt{a_x^2 + a_y^2 + a_z^2} > 12g$ with a sharp jerk transient ($\frac{da}{dt} < 5\text{ ms}$). This opens a 1.5-second posture window. 2. **Phase 2 (Posture Collapse Verification):** Core 1 integrates tri-axial angular gyroscope velocity. If net pitch/roll angular shift $\Delta\theta > 70^\circ$ within 1.5 seconds, posture collapse is confirmed. 3. **Phase 3 (Post-Impact Flaccid Atonia Variance Filter):** A 10-second sliding buffer calculates acceleration variance. If the soldier is conscious, taking cover, or crawling, muscle tremors keep variance $\text{Var}(a) > 0.05g^2$, aborting the alarm. But if voluntary motor control ceases due to concussive trauma ($\text{Var}(a) < 0.02g^2$ for 10 consecutive seconds), casualty is confirmed! 4. **Grace Window:** A 5-second acoustic buzzer and countdown alert activate; if uncancelled by the soldier, Core 0 automatically broadcasts an encrypted `PRIORITY_CASUALTY` distress beacon across the mesh with maximum +18 dBm power. In 200 combat motion trials, it achieved a **98.0% True Positive Rate with 0.0% False Alarms**!"

Slide 16: Experimental Results & Model Evaluation

SLIDE VISUAL : Normalized Confusion Matrix ($n = 108,000$), ROC Curve (AUC = 0.9585), and Hardware Performance Metrics]*

Kshitij D:

> "To validate our Edge-AI jamming detection engine under rigorous conditions, the model was evaluated against an exhaustive dataset of **108,000 synthetic and empirical RF telemetry samples** representing nominal links, barrage jamming, spot jamming, and pulse bursts.
>
> As shown on Slide 16: - **Overall Accuracy: 90.12%** - **Recall / Sensitivity (True Positive Rate): 90.74%** (correctly detecting active jamming attacks) - **Specificity (True Negative Rate): 90.00%** (clearing nominal links) - **Precision: 62.57%** (with an F1-Score of **74.07%**) - **Area Under the ROC Curve (ROC-AUC):** An outstanding **0.9585**, proving near-ideal class separability! - **Inference Latency:** Executes in **3.14 ms**, allowing real-time threat detection across incoming frames. - **Memory Footprint:** The entire 100-tree compiled model occupies only **18.4 KB of flash**, representing an ideal fit for resource-constrained microcontrollers.
>
> **Battery Autonomy:** Tested with a Fluke 287 True-RMS logging multimeter across active, receive, and sleep duty cycles (average current draw 20.13 mA), a single 3000 mAh 18650 cell delivers **126.7 hours (> 5.2 days)** of continuous mission endurance!"

Slide 17: Source of Truth — Feature Verification Matrix

SLIDE VISUAL : Verification Matrix table comparing Claims, Implementation Code, and Empirical Evidence]*

Kshitij D:

"Academic and engineering integrity are paramount in this project. On Slide 17, we present our complete **Source of Truth Verification Matrix**, mapping every core feature directly to verified code in our repository and physical bench evidence: - **LoRa CSS Burst:** Implemented in `main.cpp` using RadioLib; verified via SDR waterfall captures achieving sub-100 ms burst profiles. Status: **DEMONSTRATED**. - **Dual-Core FreeRTOS Isolation:** `xTaskCreatePinnedToCore()` separating radio and UI contexts via queues. Status: **DEMONSTRATED**. - **Authenticated AES-256-GCM:** 96-bit random IV and 128-bit GMAC tag verification executed on every frame via mbedTLS. Status: **DEMONSTRATED**. - **Asynchronous CAD-FHSS:** 15-channel hop engine written with 37-symbol strobe rendezvous math. Status: **HARDWARE TESTED**. - **Delay-Tolerant Networking (DTN):** SPIFFS flash serialization into `/dtn_XXXX.bin` with automatic queue evacuation upon heartbeat restore. Status: **IMPLEMENTED**. - **Edge-AI Electronic Warfare Detection:** 100-tree Isolation Forest running in 3.14 ms on the gateway MCU. Status: **VALIDATED**. - **Commandable Zeroization:** Two-stage slide-to-arm `CMD:ZERO` payload triggering key overwrite and flash erase. Status: **DEMONSTRATED** on physical hardware."

Slide 18: Strategic Comparative Benchmark

SLIDE VISUAL : Comprehensive Benchmark Table: S.P.E.C.T.R.E. vs Harris AN/PRC-152A, Thales MBITR2, Meshtastic, and DMR]*

Kshitij D:

"To understand where S.P.E.C.T.R.E. sits in the modern communications landscape, Slide 18 provides a direct qualitative benchmark: - **Military Tactical Radios (Harris Falcon III / Thales MBITR2):** Offer mature Type-1 encryption and frequency hopping, but cost between \$12,000 and \$22,000 USD per unit, are heavily export-controlled under US ITAR regulations, and rely on long continuous-wave voice transmissions that expose troops to RDF detection. - **Civilian Mesh Platforms (Meshtastic):** Are low-cost and open source, but rely on static pre-shared channel keys, fixed frequencies without ECCM, have no jamming anomaly awareness, and offer zero anti-tamper zeroization. - **S.P.E.C.T.R.E.:** Bridges this vital gap. It delivers authenticated AES-256-GCM encryption, sub-100 ms LPD bursts, on-chip machine learning jamming detection, delay-tolerant networking, and instant hardware zeroization — all on an open, auditable, sovereign architecture at a Bill of Materials cost of approximately **\$45 USD (under ₹3,800 INR)** per node!"

Slide 19: Technology Roadmap

SLIDE VISUAL : Phased roadmap timeline — Next (0–3 months), Mid (3–9 months), Long (9+ months)]*

Kshitij D:

> "Looking forward, we have established an actionable technological roadmap across three execution horizons: - **Near-Term (0–3 Months):** Wire the designed ephemeral SECP256R1 ECDH handshake into the default Over-The-Air frame exchange, bind headers as authenticated additional data (AAD), add packet sequence replay windows, and re-tune the FHSS band plan. - **Mid-Term (3–9 Months):** Field range trials across complex non-line-of-sight topographies, retrain the Isolation Forest on live SDR-captured interference, transition breadboards to custom 4-layer PCBs with CNC-machined aluminum enclosures, and integrate the MPU-6050 casualty hardware. - **Long-Term (9+ Months):** Migrate to Post-Quantum Cryptography (PQC) using the newly standardized NIST FIPS 203 Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM / Kyber-768), and deploy airborne tethered drone repeaters for 25+ km beyond-line-of-sight coverage."

Slide 20 & 21: Strategic Value & Alignment with National Goals

SLIDE VISUAL : Mapping to United Nations SDGs (9, 11, 16) and Atmanirbhar Bharat vision, plus Primary References]*

Kshitij D:

> "Beyond tactical military utility, S.P.E.C.T.R.E. delivers decisive societal and geopolitical value: 1. **Atmanirbhar Bharat (Sovereign Defense Self-Reliance):** Modern defense communications cannot depend on foreign proprietary radios that contain closed firmware, unknown hardware backdoors, or remote kill-switches. S.P.E.C.T.R.E. proves that sovereign, 100% auditable tactical communication security can be achieved on accessible semiconductor platforms. 2. **United Nations Sustainable Development Goals:** - **SDG 9 (Industry, Innovation & Infrastructure):** Resilient, decentralized infrastructure that operates when telecommunication backbones fail. - **SDG 11 (Sustainable Cities & Communities) & SDG 16 (Peace & Strong Institutions):** Rapid, zero-infrastructure communications for disaster relief agencies (NDRF), search-and-rescue teams during catastrophic floods or earthquakes, and border security forces."

Slide 22: Conclusion & Technical Demonstration

SLIDE VISUAL : Final Evaluation Slide with key quantitative metrics, project contributors, and invitation for Q&A and Live Demo]*

Kshitij D:

> "To conclude: S.P.E.C.T.R.E. successfully demonstrates that by combining Low Probability of Detection LoRa burst chirps, zero-trust authenticated encryption, dual-core FreeRTOS task isolation, delay-tolerant flash networking, and on-chip Edge-AI inference, we can create an impenetrable, off-grid communication lifeline for those operating on the front lines and in disaster zones.
>
> We express our heartfelt gratitude to our guide Dr. Arathi R Shankar ma'am, our HOD Dr. Beena Ullala Mata ma'am, and the faculty of the Department of Electronics and Communication Engineering for their invaluable mentorship.
>
> Our physical testbed — comprising our deployed field terminals, headless C2 gateway, and live Tactical Command Center dashboard — is powered on and ready.
>
> Thank you. We are now open for technical questions and would be delighted to demonstrate the live system."

End of Script. Total estimated delivery time: 14 to 18 minutes (approx. 3.5 to 4.5 minutes per speaker).